
NBA Draft Value & Decision Analysis (1990-2020)

Identifying Value and Improving Late Draft Decisions

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Why the NBA Draft Matters

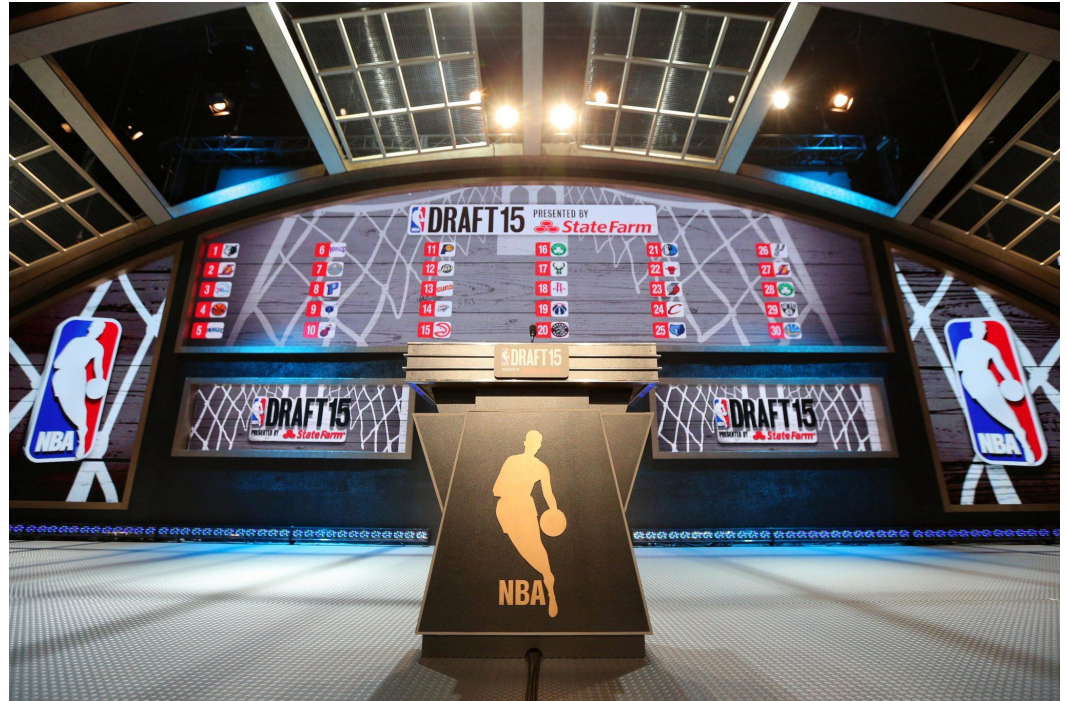
- Each year, teams invest heavily in player acquisition
 - Millions spent on scouting & player evaluation
 - Top picks alone can cost \$60M+ per player
- But outcomes of players vary drastically
 - Second round picks often sign non-guaranteed contracts
 - 70% of late second-round picks produce close to no value
- High investment does NOT guarantee success
- Average Length of an NBA career is 4 - 5 years (NBA.com)

Research Objective

Teams invest heavily in draft picks, but outcomes are highly uncertain

This leads to two key questions:

- Where is Draft Value Actually Concentrated?
- Is the second round truly a game of chance, or are there identifiable patterns of success?



Data & Approach

Dataset (NBA Drafts: 1990-2020)

- 1,800+ players across 30 years
- Includes player attributes & career outcomes
- Scraped, cleaned, and structured from Basketball Reference

Analytical Framework

- Measure success
 - Career length
 - Long-term value (win shares)
- Compare across draft tiers
 - Top 3, lottery, non-lottery 1st, early 2nd, and late 2nd
- Predictive Modeling
 - Logistic regression
 - ROC/AUC evaluation
 - Probability of a player being a bust or success

Draft Tier Structure

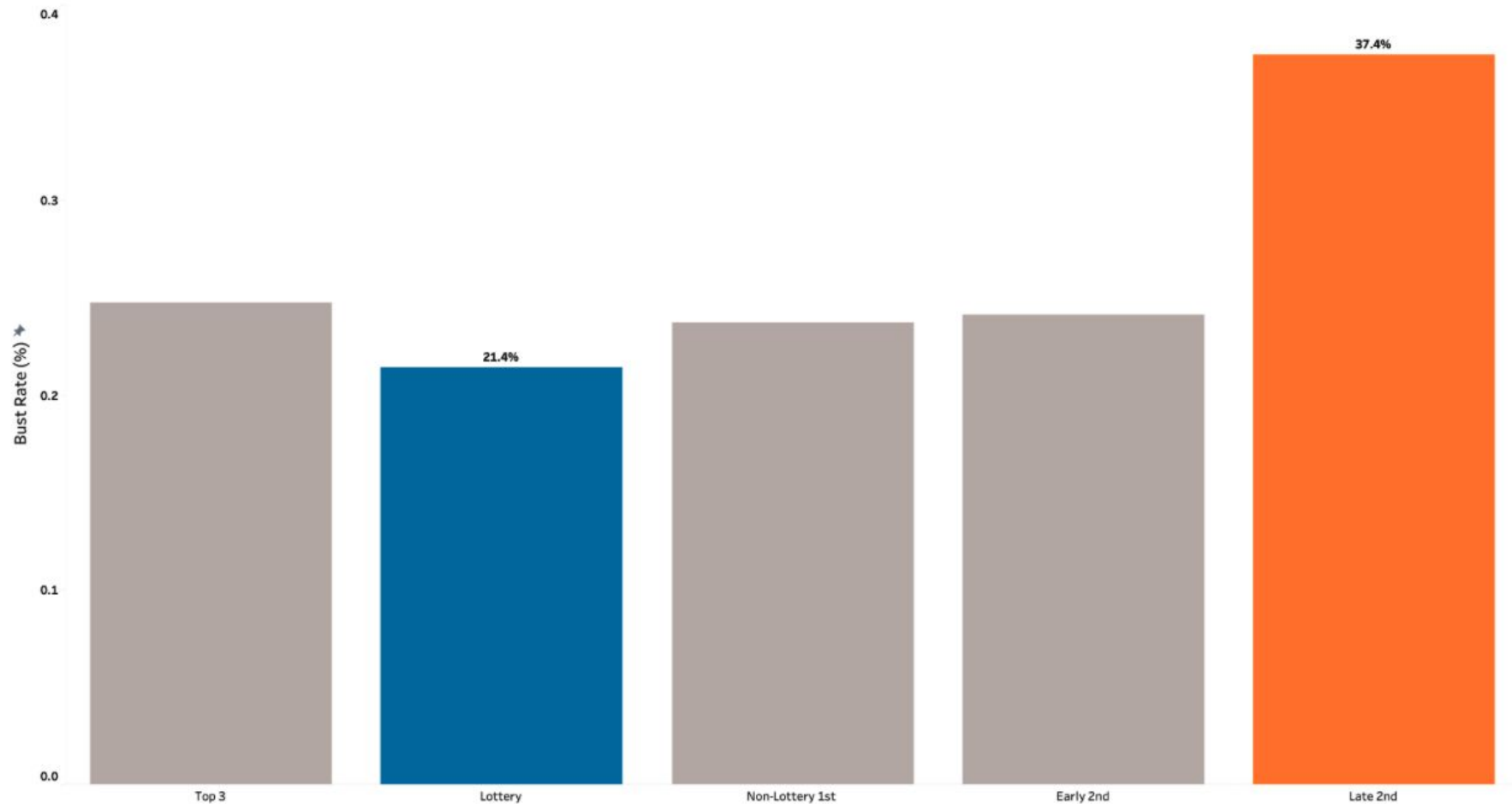
- Draft picks grouped into performance tiers
- Each tier is evaluated relative to its own expectations
- Allows for fair comparison across draft positions

Tier	Picks	Avg WS	Bust Rate
Top 3	1-3	60.7	24.7%
Lottery	4-14	33.9	21.4%
Non-Lottery 1st	15-30	18.5	23.7%
Early 2nd	31-45	9	24.1%
Late 2nd	46-60	4.3	37.4%

Bust = players below tier-specific thresholds based on Win Shares and career length, excluding All-Stars and high-performing careers.

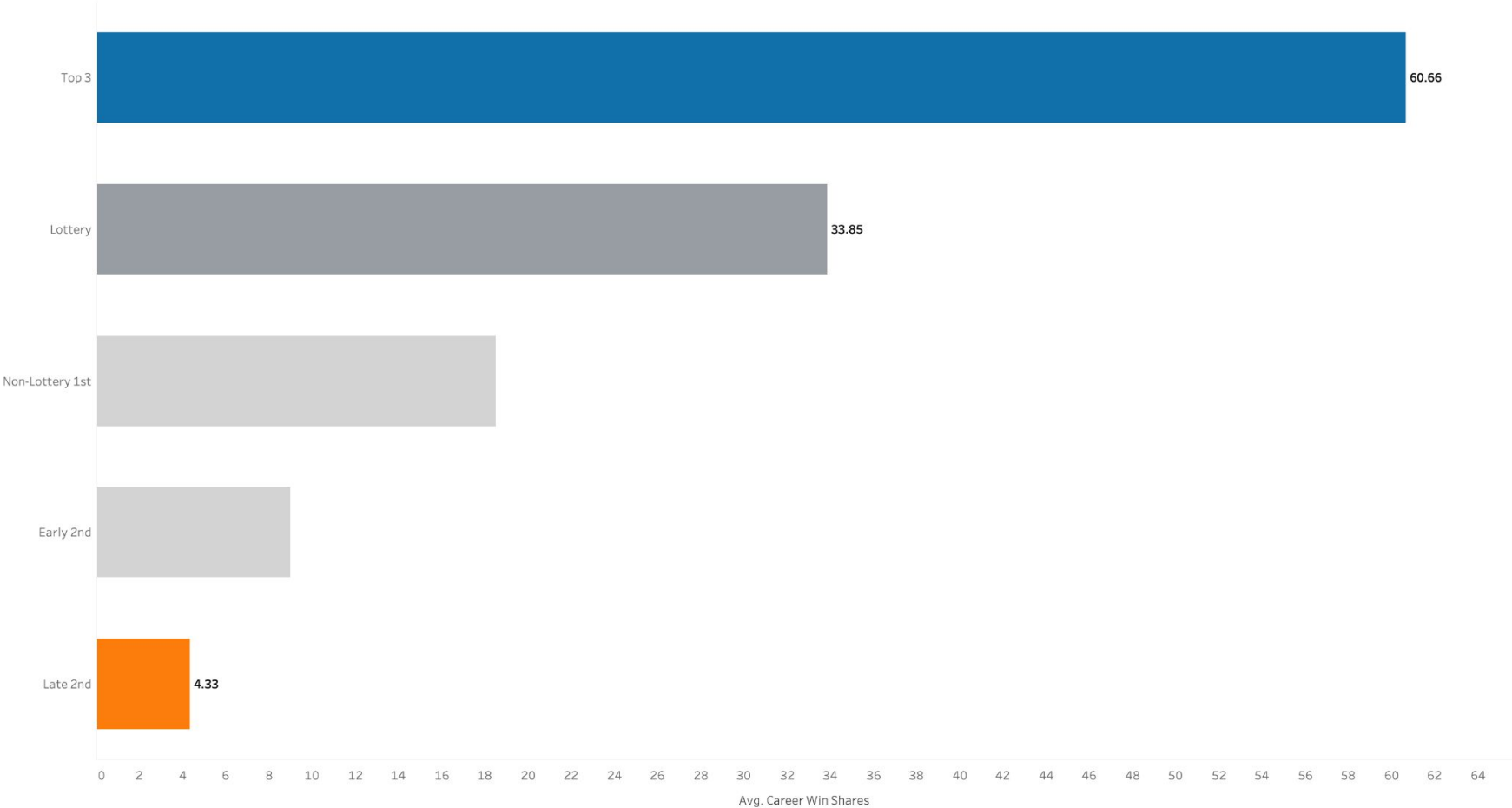
Bust Rate By Draft Tier

Tier-adjusted risk highlights late 2nd round volatility



Average Career Value by Draft Tier

Draft value declines sharply outside top selections



Key Takeaways

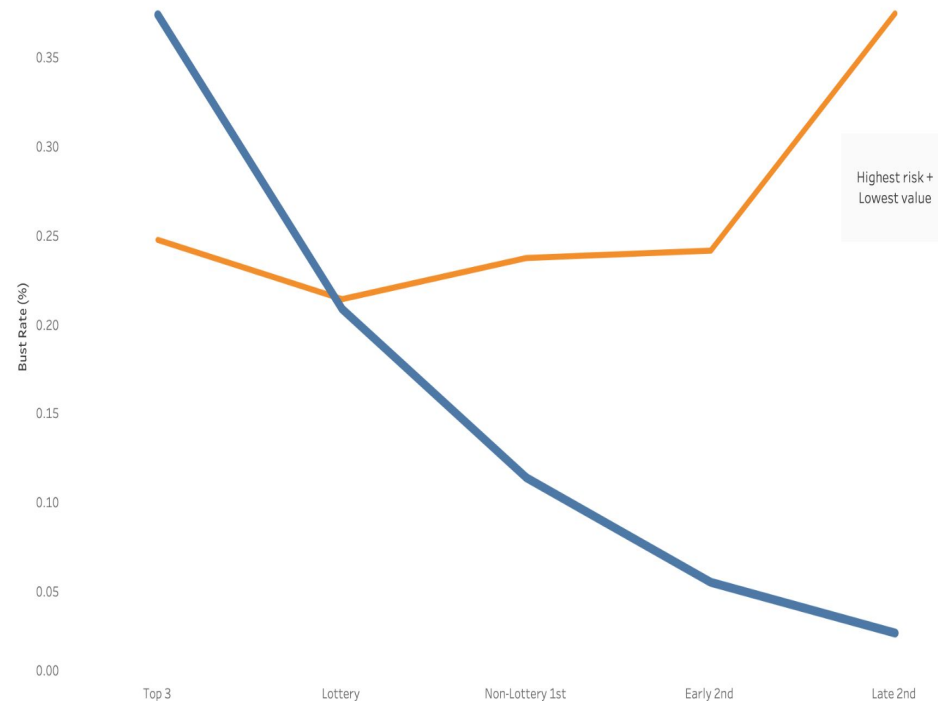
Value declines - but risk tells a different story

- Higher picks buy upside, not safety
- Late 1st and early 2nd round picks have similar risk profiles
- Late 2nd round is the main tier with clearly worst outcomes

Measure Names

Avg. Bust Zero Filled
Avg. SCALED WS

Value Vs. Risk Across Draft Tiers



Is There Signal In the Chaos?

Finding Value in the Second Round

What Does the Second Round Look Like

906 Second-Round Players

- Early 2nd: 460 | Late 2nd: 446

Median Outcome

- Early 2nd: 0.2 WS | Late 2nd: 0.0

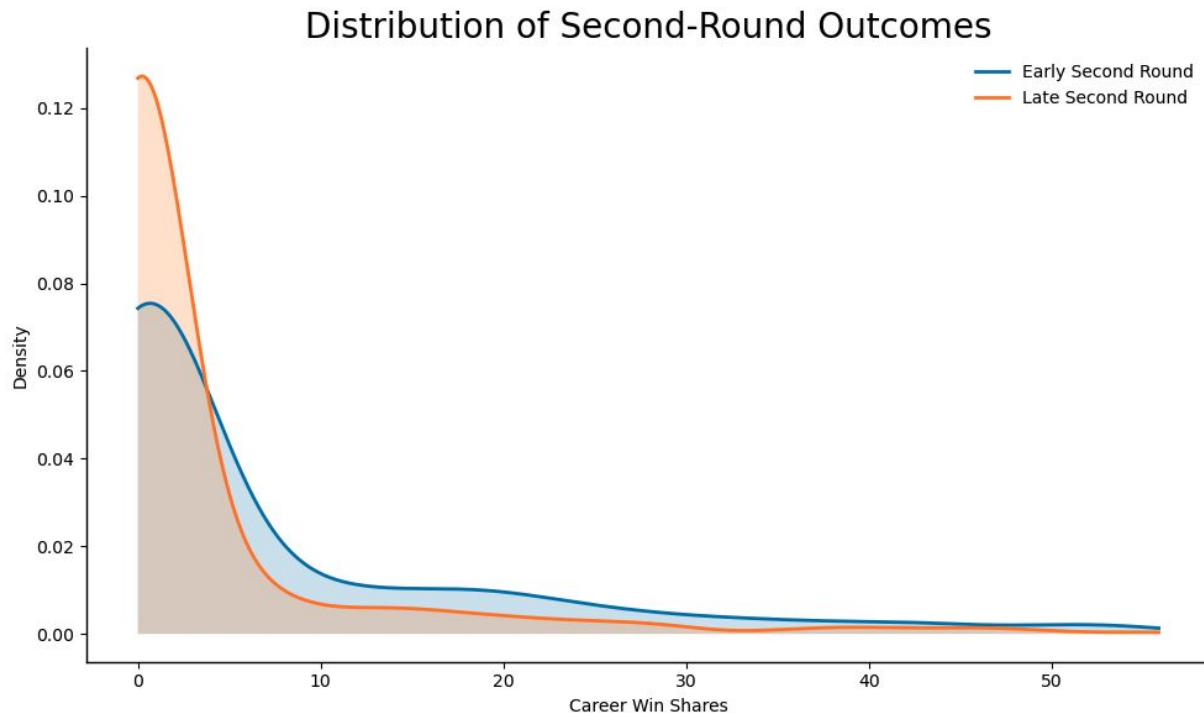
Key Insight

- Most players provide little to no value
- A small number of players drive almost all outcomes

Where Does Signal Exist in the Second Round

Probability of finding a “Steal”

- Overall: 15.8%
- Early 2nd: **21.5%**
- Late 2nd: **9.9%**



Where Does Signal Exist in the Second Round Cont.

What This Means

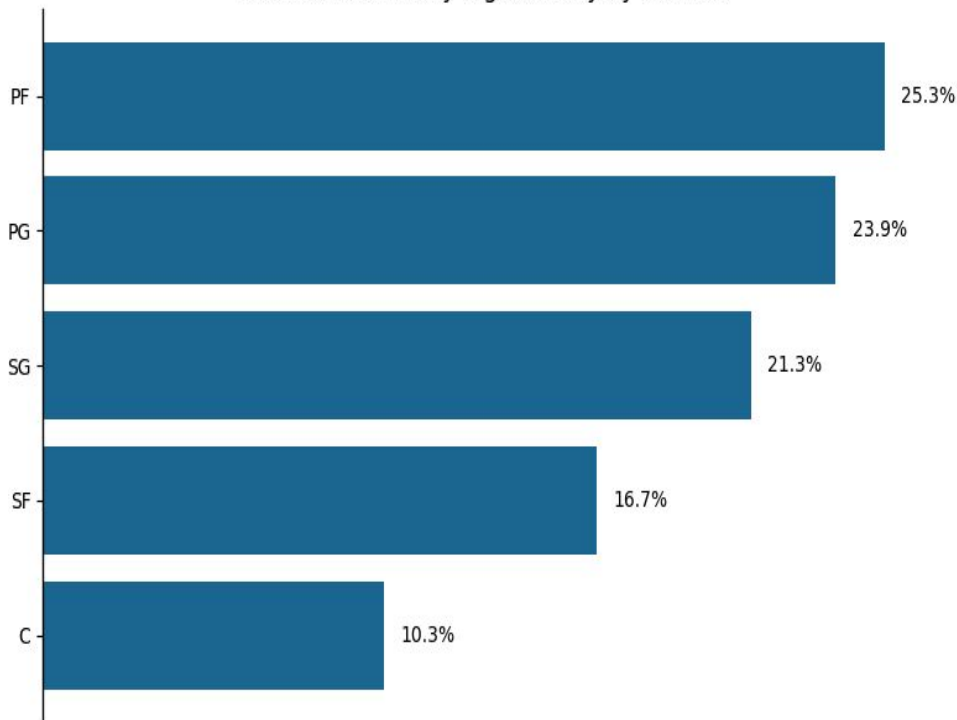
- Success Probability drops by over 50% from early to late second round
- Early 2nd shows a real measurable signal
- Late 2nd outcomes are less predictable

Draft Implications

- Early Picks - higher hit rate, lower variance
- Late Picks - lower hit rate, higher variance
- The second round is not random, but the signal fades quickly

Where Does the Signal Come From?

Success Rates Vary Significantly by Position



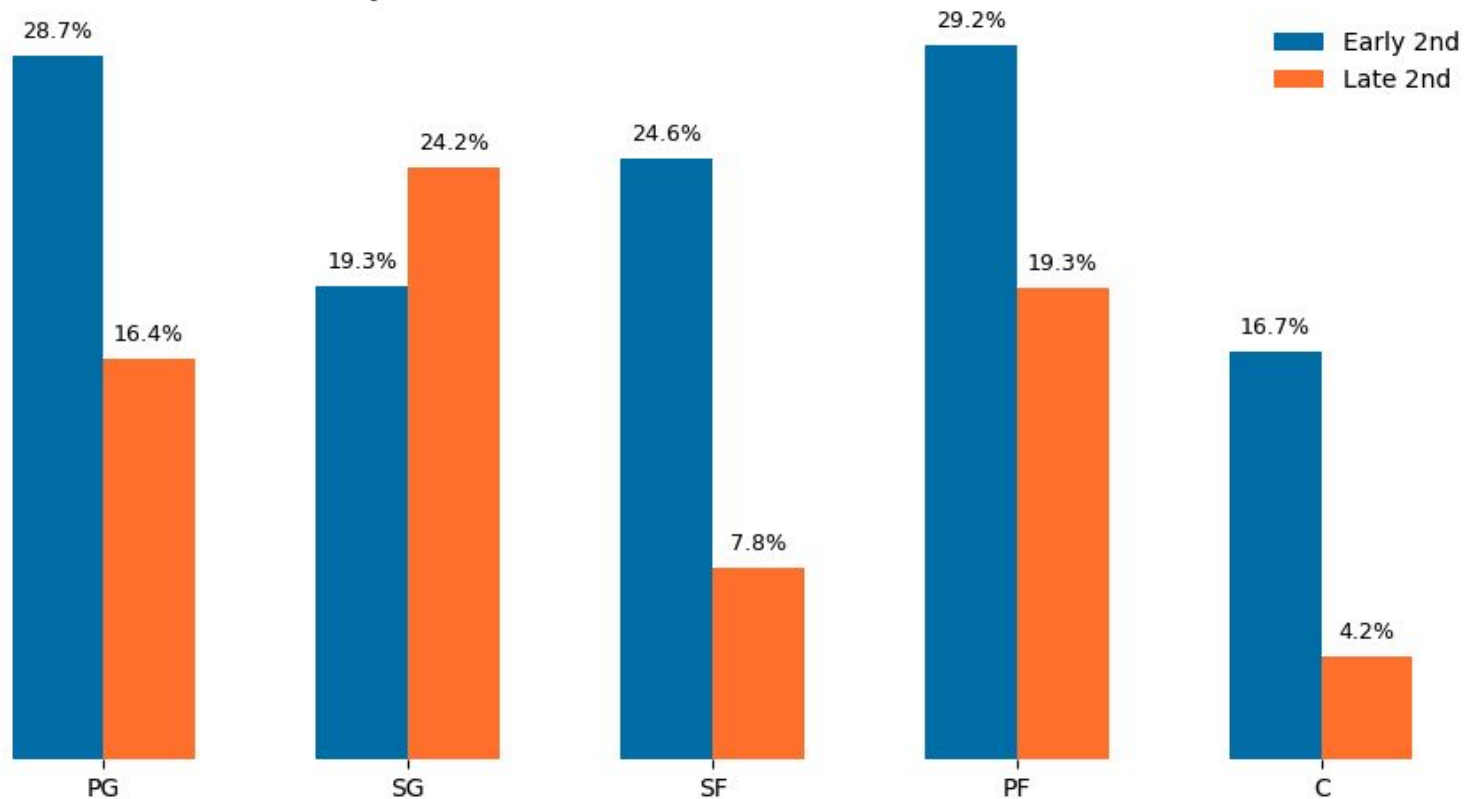
Key Findings

- Forwards and guards outperform centers in the second round
- Power forwards have the highest success rate
- Centers have the lowest hit rate and lowest median outcomes

What this means

- Position is a measurable edge in the second round
- Not all positions carry equal draft value
- Despite highly variable outcomes, positional structure still exists

How Does Position Change Steal Probability Across the Second Round?

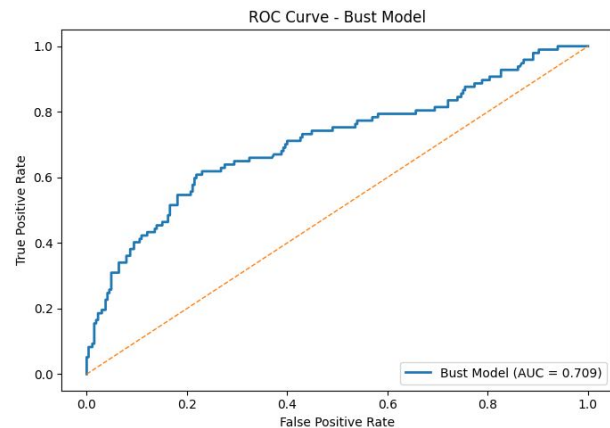
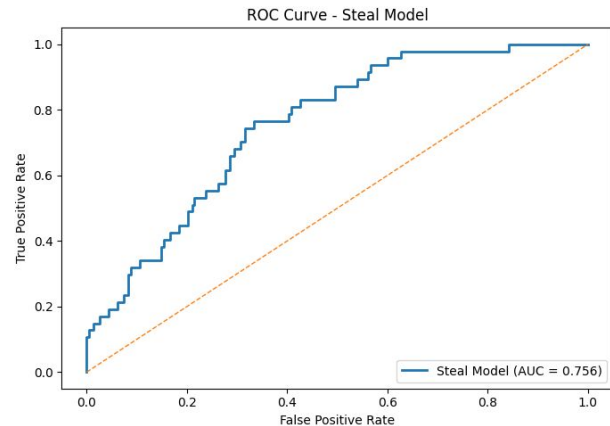


From Patterns to Predictions

Estimating risk and upside before draft night

Predictive Modeling: Binary Logistic Regression

- Binary classification using logistic regression
- Outcomes:
 - Steal (0/1)
 - Bust (0/1)
- Focused on non-lottery and second-round picks
- Evaluated with ROC/AUC
 - Steal AUC: 0.76
 - Bust AUC: 0.71
- Binary predictions create conflicting decisions
 - Players can show high risk AND high upside



Multiclass Model for Draft Risk Assessment

- Reframed as a multiclass logistic model
- Outcomes:
 - Bust
 - Steal
 - Neither
- Produces a full probability distribution
- Enables risk vs. reward evaluation
- Supports draft decision-making under uncertainty
- Designed to estimate risk profiles, not to force a single classification

Interactive Draft Risk Calculator

- Input player profile characteristics
- Returns bust, steal, and baseline probabilities
- Summarizes risk-reward profile
- Designed for real-time draft decision support

PLAYER INPUTS

Draft Tier	Age
Non-Lottery 1st	21
Position	International
PG	Yes
Height (cm)	Weight (lbs)
218.1	299
Shoots	Power School
Left	No

Power School is disabled for international players.

CalculateGenerate Profile

Interactive Draft Calculator

Built from historical NBA draft data (1990 - 2020)

Key Takeaways

- Draft value is concentrated early, but meaningful upside still exists outside the lottery
- The second round is not random - risk can be evaluated through identifiable player profiles
- This model quantifies draft risk, translating uncertainty into a practical decision tool

Thank You

Questions?

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